

Kuanwei Chen

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Summary

M.S. Computer Science student at National Central University specializing in Artificial Intelligence, with research presented at three international conferences. Experienced in building computer vision and NLP systems for scene text recognition, sign language translation, and biometric recognition using PyTorch and TensorFlow. Seeking AI/ML engineering or computer vision research roles.

Education

National Central University | Master, Computer Science, Artificial Intelligence Taiwan 2025 – Present
National Changhua University of Education | Bachelor, Electrical Engineering, (GPA 3.8/4.0) Taiwan 2021 – 2025

Experience

National Changhua University of Education, AI Application Lab | *Project Assistant* Jun 2024 – Dec 2024

- Collaborated with 8 international interns under the TEEP and IIPP research programs, delivering 3 multimodal AI prototypes in biometric and action recognition using PyTorch and TensorFlow, contributing to a peer-reviewed publication at IET ICETA.
- Built Pandas and NumPy-based preprocessing, dataset, and evaluation pipelines that streamlined model experimentation and validation across computer vision and speech-related research tasks.

Publication

1. **K. Chen**, K. Zheng, and M. Tsai, “STRLite: Lightweight Masked Autoencoders for Adaptable Scene Text Recognition,” IEEE ICCE-TW (accepted), 2026.
2. **K. Chen** and T. Lin, “SignDATA: Data Pipeline for Sign Language Translation,” arXiv:2604.20357, 2026.
3. **K. Chen** and M. Tsai, “Towards Compact Sign Language Translation: Frame Rate and Model Size Trade-offs,” IEEE ICCE-TW (accepted), 2026.
4. **K. W. Chen**, T. Y. Lin, W. R. Yang, A. Kesarwani, and R. Singh, “Two-step Authentication: Multi-biometric System Using Voice and Facial Recognition,” IET Conf. Proc., 2024, doi: 10.1049/icp.2024.4141.

Projects

STRLite: Lightweight Masked Autoencoders for Adaptable Scene Text Recognition | github.com/balaboom123/STRLite

- Co-first author; designed a 6M-parameter Vision Transformer (ViT) with Masked Autoencoder (MAE) pretraining and fine-tuning with an autoregressive decoder for Scene Text Recognition.
- Achieved 93.82% weighted accuracy on six standard Computer Vision benchmarks and 81.03% on U14M.

SignDATA: Data Pipeline for Sign Language Translation | github.com/balaboom123/signdata-slt

- First author; open-source repo with 100+ stars; built a config-driven data preprocessing, and dataset management system for standardizing heterogeneous sign-language corpora for AI/ML Pipeline development.
- Designed processing pipeline integrating YOLO/MMDet detection and MediaPipe/MMPose landmark extraction for structured feature engineering, applying system-design principles for multi-dataset scalability.

Towards Compact Sign Language Translation: Frame Rate and Model Size Trade-offs

- First author; built a 77M-parameter end-to-end Natural Language Processing (NLP) pipeline.
- Reduced computation by ~75% with minimal BLEU impact (10.06 to 9.53), providing efficiency trade-off insights.

Two-step Authentication: Multi-biometric System Using Voice and Facial Recognition |

github.com/NCUE-EE-AIAL/Two-Step-Muti-Biometric-Authentication-System

- First author; led a team of 2 interns in designing a multi-modal AI/ML system combining Machine Vision and Audio Recognition/Speaker Verification.
- Designed a ResNet-based Convolutional Neural Network (CNN) with Fbank feature extraction and triplet loss for voice verification (99.1%); fine-tuned VGG16 with MTCNN object detection in TensorFlow (95.1%).

SEO Content Intelligence Engine | github.com/balaboom123/SEO-Content-Intelligence-Engine

- Built a full-stack Next.js web app in TypeScript that scrapes top-10 Google SERP results, extracts NLP entities via OpenAI, and clusters them to reverse-engineer search intent for SEO competitor analysis.
- Implemented Supabase authentication with SQL Row-Level Security policies, interactive Recharts visualizations, Google Sheets export, and deployed on Vercel with rate limiting and API security hardening.

Skills

- **Programming & Tools:** Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, Git, Linux, Docker
- **ML Frameworks & Stack:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV, NumPy, Pandas, YOLO, MMDet, MediaPipe
- **Web Development:** React, Next.js, Vue, Supabase, Vercel
- **Languages:** English (IELTS 6.5, TOEIC 830), Mandarin (native), Hokkien (native)